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FGOOPNet: A Fuzzy Deep Learning Approach for Mammographic Breast Cancer Classification

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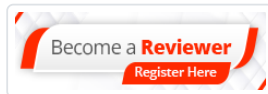
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Abstract

Introduction: Breast cancer continues to threaten millions of women globally, and early detection is essential for saving lives. However, inconsistencies in radiological interpretation and uncertainty in lesion assessment limit diagnostic reliability. This study proposes FGOOPNet, a fuzzy deep ensemble model designed to enhance mammographic breast cancer classification accuracy and decision transparency.

Methods: Seven complementary CNN architectures, InceptionV3, ResNet-50V2, ResNet-152, DenseNet-121, DenseNet-201, VGG19, and Xception, were integrated into a unified ensemble framework using the Geometrically Optimum and Online Weighted Ensemble (GOOWE) strategy. GOOWE computes classifier weights through a geometrically optimal least-squares projection, assigning stronger influence to models with lower validation errors. To further refine decision boundaries and quantify predictive uncertainty, a Gaussian-based fuzzy inference module was applied to the ensemble outputs. The proposed system was evaluated on a hybrid dataset combining the public VinDr-Mammo database with an expert-annotated private cohort from regional hospitals, with performance assessed using accuracy, F1-score, and robustness under multiple experimental settings.

Results: FGOOPNet achieved 98.4% accuracy and highly balanced F1-scores for both benign and malignant classes, outperforming individual CNN models and conventional ensemble techniques. The fuzzy inference layer improved reliability in borderline cases by filtering low-confidence predictions and reducing misclassification.

Discussion: The results demonstrate that combining GOOWE weighting with fuzzy Gaussian reasoning enhances both predictive performance and interpretability. This uncertainty-aware mechanism helps address limitations in conventional CAD systems, particularly in heterogeneous breast tissue and visually ambiguous regions.

Conclusion: FGOOPNet offers a robust and explainable solution for mammographic breast cancer classification, highlighting its potential for integration into clinical decision-support workflows. Future work will focus on extending the model to multi-modal imaging and large-scale multi-center validation.

Keywords: [Breast cancer](#), [Ensemble learning](#), [GOOWE](#), [Fuzzy Gaussian membership](#), [Mammography](#), [Deep learning](#), [CAD systems](#)

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